UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO. | CONFIRMATION NO. |
|---|---|---|---|---|
| 12/890,234 | 09/24/2010 | John R. Andrews | 1776-0364 | 7918 |

76360 7590 07/30/2012

MAGINOT, MOORE & BECK LLP
111 MONUMENT CIRCLE
SUITE 3250
INDIANAPOLIS, IN 46204

| EXAMINER | |
|---|---|
| JACKSON, JUANITA DIONNE | |
| ART UNIT | PAPER NUMBER |
| 2853 | |

| MAIL DATE | DELIVERY MODE |
|---|---|
| 07/30/2012 | PAPER |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

| Office Action Summary | Application No. | Applicant(s) | |
|---|---|---|---|
| | 12/890,234 | ANDREWS ET AL. | |
| | Examiner | Art Unit | |
| | JUANITA D. JACKSON | 2853 | |

*-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --*

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE *3* MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1)☒ Responsive to communication(s) filed on *Election filed 6/29/2012*.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims**

5)☒ Claim(s) *1-19* is/are pending in the application.

5a) Of the above claim(s) *1-6* is/are withdrawn from consideration.

6)☐ Claim(s) _____ is/are allowed.

7)☒ Claim(s) *7-19* is/are rejected.

8)☐ Claim(s) _____ is/are objected to.

9)☐ Claim(s) _____ are subject to restriction and/or election requirement.

**Application Papers**

10)☐ The specification is objected to by the Examiner.

11)☒ The drawing(s) filed on *24 September 2010* is/are: a)☒ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

12)☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

**Priority under 35 U.S.C. § 119**

13)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

a)☐ All b)☐ Some * c)☐ None of:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☒ Notice of References Cited (PTO-892)
2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948)
3) ☒ Information Disclosure Statement(s) (PTO/SB/08) Paper No(s)/Mail Date *9/24/2010*.
4) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____ .
5) ☐ Notice of Informal Patent Application
6) ☐ Other: _____.

## DETAILED ACTION

### *Election/Restrictions*

1. Applicant's election without traverse of Group II, claims 7-19 in the reply filed on 6/29/2012 is acknowledged.

2. Claims 1-6 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected invention there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 6/29/2012.

### *Information Disclosure Statement*

3. The information disclosure statement (IDS) submitted on 9/24/2010 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

### *Claim Rejections - 35 USC § 112*

4. The following is a quotation of the second paragraph of 35 U.S.C. 112:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

5. Claims 7-19 are rejected under 35 U.S.C. 112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which applicant regards as the invention.

In claim 7, line 8 the recitation of "each actuator in each array of actuators" has no antecedent basis.

In claims 14 and 19, line 2 the recitation of "the top surfaces of the walls" has no antecedent basis.

In claim 15, it is not clearly understood what the difference is between "a flexible member" recited in line 6 and "a membrane" recited in line 9.

***Claim Rejections - 35 USC § 102***

6. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

7. Claims 7-9 and 11 are rejected under 35 U.S.C. 102(b) as being anticipated by Barnes et al. (US 7,254,890 B2).

Barnes et al. discloses a multi-layered structure (Fig. 4A) suitable for use in manufacturing inkjet printheads comprising: **1)** a wafer (34) including a plurality of die sites (14) arranged in a predetermined pattern (col 5, lns 9-10), a polymer layer (46) including a plurality of apertures (52) arrays arranged in a predetermined pattern corresponding to the predetermined pattern of the plurality of die sites (14) on the wafer, the polymer layer (46) and wafer being bonded together to enable each actuator (heater resistors 36) in each array of actuators formed in each die site (14) on the wafer (34) to be aligned with an aperture (52) in a corresponding aperture array formed in the polymer layer (46) (col 5, lns 27-31; col 6, lns 29-36), **2)** wherein each actuator (36) in each array of actuators is positioned to urge ink through a corresponding aperture (52) in each array of apertures (as shown in Fig. 4B), **3)** the wafer (34) being primarily composed of silicon (col 5, lns 27-28), and **4)** an inter-die region that separates the die

sites (14) in the plurality of die sites on the wafer (34) from one another; and a corresponding region that separates the aperture arrays in the plurality of aperture arrays in the polymer layer from one another (well known). Barnes et al. further at least teaches that actuators may be piezoelectric devices or heater resistors (col 5, lns 27-30)

***Claim Rejections - 35 USC § 103***

8. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

> (a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negatived by the manner in which the invention was made.

9. Claims 10 and 12-19 rejected under 35 U.S.C. 103(a) as being unpatentable over Barnes et al. (US 7,254,890 B2) in view of Shingyohuchi (US 6,811,238 B2).

Barnes et al. discloses a multi-layered structure (Fig. 4A) suitable for use in manufacturing inkjet printheads comprising: **1)** a wafer (34) including a plurality of die sites (14) arranged in a predetermined pattern (col 5, lns 9-10), a polymer layer (46) including a plurality of apertures (52) arrays arranged in a predetermined pattern corresponding to the predetermined pattern of the plurality of die sites (14) on the wafer, the polymer layer (46) and wafer being bonded together to enable each actuator (36) in each array of actuators formed in each die site (14) on the wafer (34) to be aligned with an aperture (52) in a corresponding aperture array formed in the polymer layer (46) (col 5, lns 27-31; col 6, lns 29-36), **2)** wherein each actuator (36) in each array of actuators is positioned to urge ink through a corresponding aperture (52) in each array of

apertures (as shown in Fig. 4B), **3)** the wafer (34) being primarily composed of silicon (col 5, lns 27-28), **4)** an inter-die region that separates the die sites (14) in the plurality of die sites on the wafer (34) from one another; and a corresponding region that separates the aperture arrays in the plurality of aperture arrays in the polymer layer from one another (well known), and **5)** a polymer layer (46) including a plurality of aperture arrays arranged in a predetermined pattern corresponding to the predetermined pattern of the plurality of die sites (14) on the wafer (34), each aperture array on the polymer layer having a plurality of apertures (52) with each aperture placed in fluid communication with a corresponding fluid chamber (48) in each die site on the wafer, and the polymer layer and wafer being bonded together (col 6, lns 31-36 as shown in Fig. 4A). Barnes et al. further at least teaches that actuators may be piezoelectric devices or heater resistors (col 5, lns 27-30).

Barnes et al. does not teach **1)** each actuator further comprising a region of polysilicon on the wafer, **2)** an array of actuators formed on a surface of each die site on the wafer, each actuator having a flexible membrane, and a plurality of walls extending from the surface of each die site on the wafer to a position above the flexible membranes, the walls and flexible membranes forming a plurality of fluid cavities, with a fluid cavity positioned over each actuator in each of the arrays of actuators, **3)** wherein the polymer layer is bonded to top surface of the walls, **4)** an adhesive layer positioned between the top surfaces of the walls and the polymer layer, **5)** an array of fluid chambers positioned over each array of actuators in each die site on the wafer, each fluid chamber in the array of fluid chambers being formed above a membrane in the

array of actuators. Shingyohuchi at least teaches each actuator further comprising a region of polysilicon on the wafer (well known), an array of actuators (diaphragm 50/electrode 55) formed on a surface of each die site (42) on the wafer, each actuator having a flexible membrane (diaphragm 50) (col 8, lns 52-57), and a plurality of walls (formed by 41) extending from the surface of each die site on the wafer to a position above the flexible membranes (50), the walls (41) and flexible membranes forming a plurality of fluid cavities (gap 56), with a fluid cavity (56) positioned over each actuator (50/55) in each of the arrays of actuators (col 8, lns 52-57 as shown in Fig. 5), wherein the polymer layer (nozzle plate 43) is bonded to top surface of the walls (41) (as shown in Fig. 5), an adhesive layer positioned between the top surfaces of the walls and the polymer layer (43) (col 9, lns 54-55), an array of fluid chambers (46) positioned over each array of actuators (50/55) in each die site (42) on the wafer, each fluid chamber (46) in the array of fluid chambers being formed above a membrane in the array of actuators (col 8, lns 5-12, lns 51-57). It would have been obvious at the time the invention was made to a person having ordinary skill in the inkjet art to substitute the micro ejection head utilizing heater resistor actuators of Barnes et al. with the ink jet head utilizing piezoelectric actuators as taught to be old by Shingyohuchi for the purpose of ejecting ink droplets of a considerable size and a finer size with respect to a nozzle diameter.

## ***Conclusion***

10. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

The prior art of Yamanaka et al. (US 6,568,794 B2) discloses actuators (oscillation plate 410/electrode 415)comprising a region (formed by electrode 415) of polysilicon on the wafer (cl 12, lns 22-23).

**Contact Information**

11. Any inquiry concerning this communication or earlier communications from the examiner should be directed to JUANITA D. JACKSON whose telephone number is (571)272-2153. The examiner can normally be reached on Flex (Monday-Friday 9:00 am -5:30 pm).

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Stephen Meier can be reached on (571) 272-2149. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a

USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

JUANITA D. JACKSON
Primary Examiner
Art Unit 2853

/JUANITA D. JACKSON/
Primary Examiner, Art Unit 2853
July 25, 2012